# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME: CSA07 COURSE CODE: Computer Networks**

**COURSE OUTCOME COVERED**

**CO1:** Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

**CONCEPT MAPPING**

*Code of Conduct:*

*I* K. Mohammed Tasin *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

*Signature of the student:*

| Criteria | Weightage (%) | Q1 10marks | Q2 10marks | Q3 10marks |
|---|---|---|---|---|
| Concept Identification | 20% | 1 | 1 | 1 |
| Concept Mapping Structure | 25% | 2 | 2 | 2 |
| Linking & Relationships | 20% | 2 | 2 | 1 |
| Application & Analysis | 20% | 2 | 2 | 2 |
| Explanation & Clarity | 15% | 1 | 1 | 1 |

**Total Marks** 23 **/30**

***Signature of the Faculty***

**CONCEPT MAPPING QUESTIONS**

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.
Analyze the missing layers and explain how their functions are essential for reliable communication.

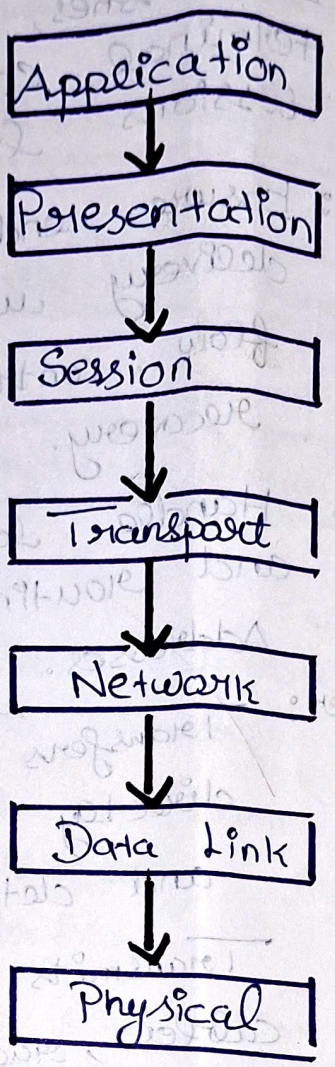
Name: K. Mohammed Jaisim
Reg No: 192260074

CO1 AT-2

Answer:-

1. Missing OSI layers and their Functions

Concept Map:-




23/30

**Explanation:-**

The OSI model consist of seven layer that work together to ensure reliable communication between devices.

1. Application layer: Provides network service to users, such as web browsing and email.

2. presentation layer: Convert data formats and performs encryption and compression.

3. Session layer: Establishes, manages, and terminal communication sessions between devices.

4. Transport layer: Ensure end-to-end data delivery using segmentation, flow control, and error recovery.

5. Network layer: Handle logical addressing and routing using IP Addresses.

6. Data link layer: Transfers frames between directly connected devices and detect transmission errors.

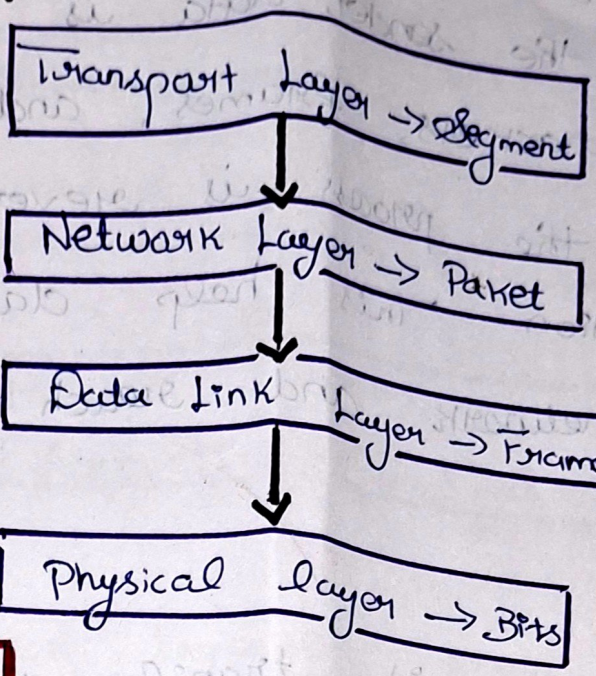
7. Physical Layers: Transmits raw bits through cables, radio waves, or other physical media.

**Conclusion:-**

All seven layers work together to provide efficient and reliable data communication.

## ②. Data units and their Respective OSI Layers

Concept Map:-




**Explanation:-**

1). Segment :- The Transport Layer divides data into smaller unit for transmission.

2). Packet :- The Network layer adds IP addressing information to support routing.

3). Frame :- The Data Link layer adds MAC addressing and error-detection information for local delivery.

4). Bits :- The physical layer converts frames into binary bits and transmission them communication medium.

How transformation support tramission:-

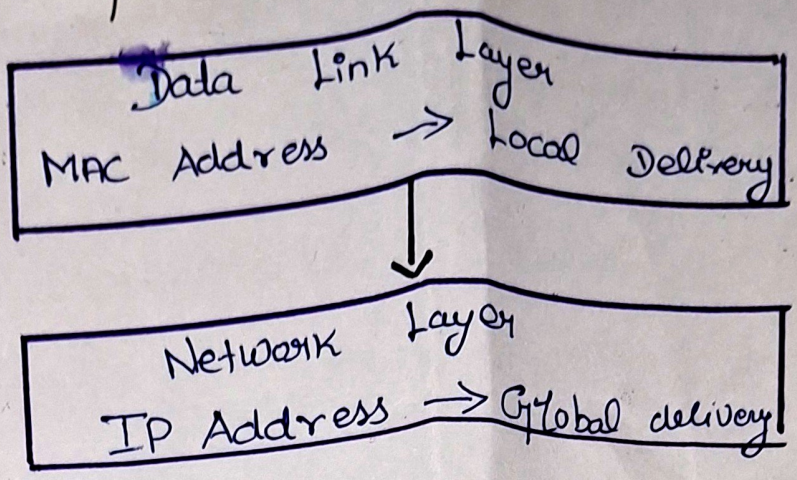
At the sender, data is encapsulated into segments, packets, frames, and bits. At the receiver, the process is reversed through decapsulation. This help data travels across network and reach the correct destination.

**Conclusion:-**

Data unit transformation enables organized, efficient, and reliable communication.

MAC Address, IP Address, and Data, delivery:-

Concept Map:-




Explanation:-

1. Data Link Layer:-

Uses MAC addresses to delivery frames between devices to the same local network.

2. Network layer :-

uses IP Addresses to route Packets between different networks.

3. MAC Address :-

Identifies a network interface for Communication on the local network.

4. IP Address:-

Identifies a devices network - layer destination and helps